

Deep Generative Models

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Problem Set 3

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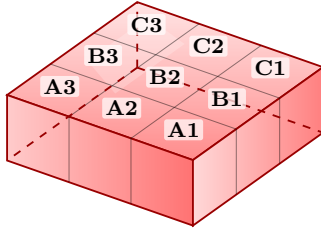
Problem 1

A: Jacobian and Log-Determinant

From the definition of the Jacobian we have :

$$\mathbf{J}_{(\mathbf{F})}(\mathbf{x}_{i,j}) = \begin{bmatrix} \frac{\partial y_{1,i,j}}{\partial x_{1,i,j}} & \frac{\partial y_{1,i,j}}{\partial x_{2,i,j}} & \cdots & \frac{\partial y_{1,i,j}}{\partial x_{c,i,j}} \\ \frac{\partial y_{2,i,j}}{\partial x_{1,i,j}} & \frac{\partial y_{2,i,j}}{\partial x_{2,i,j}} & \cdots & \frac{\partial y_{2,i,j}}{\partial x_{c,i,j}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial y_{c,i,j}}{\partial x_{1,i,j}} & \frac{\partial y_{c,i,j}}{\partial x_{2,i,j}} & \cdots & \frac{\partial y_{c,i,j}}{\partial x_{c,i,j}} \end{bmatrix}_{C \times C}$$

$$\Rightarrow \mathbf{J}_{(\mathbf{F})}(\mathbf{x}_{i,j}) = \mathbf{W}$$



Consider the tensor on left. We have three channels: 1, 2, and 3. and three pixels: A, B, and C. For pixel A we have :

$$Y_{1,1} = W_{3 \times 3} A_{:,1,1} = \begin{bmatrix} W_{11}A_1 + W_{12}A_2 + W_{13}A_3 = Y_{A_1} \\ W_{21}A_1 + W_{22}A_2 + W_{23}A_3 = Y_{A_2} \\ W_{31}A_1 + W_{32}A_2 + W_{33}A_3 = Y_{A_3} \end{bmatrix}$$

Just like pixel A, we do the same for pixels B and C. After that we stack all the output on top of each other to get :

$$Y = [Y_{A_1} \ Y_{A_2} \ Y_{A_3} \ Y_{B_1} \ Y_{B_2} \ Y_{B_3} \ Y_{C_1} \ Y_{C_2} \ Y_{C_3}]^\top$$

$$\mathbf{J} = \left[\frac{\partial Y}{\partial X} \right]_{9 \times 9}$$

The reason that the dimension is 9, is because $C \times H \times W = 3 \times 3 \times 1 = 9$ in this example and that we have 9 inputs and 9 outputs. We see that only the derivative of an output is only non-zero w.r.t to it's corresponding input pixel. so we have :

$$\mathbf{J}_{\mathbf{F}}(\mathbf{X}) = \begin{bmatrix} \underbrace{W}_{w.r.t.A} & \underbrace{0}_{w.r.t.B} & 0 \\ 0 & W & 0 \\ 0 & 0 & W \end{bmatrix}_{9 \times 9}$$

If we want to have an invertible linear transformation (such as the one we have here), the multiplication matrix should be full-rank meaning that it's det W shouldn't be zero.

Since this transformation is just a linear bijection on every spatial location of the image, assuming that W is full-rank, then the inverse transformation is :

$$W^{-1}W X_{:,i,j} = W^{-1}Y_{:,i,j} \Rightarrow X_{:,i,j} = W^{-1}Y_{:,i,j}$$

B: LU Parametrization

Based on the given parametrization, we have :

$$W = PL(U + \text{diag}(S)) \Rightarrow \det W = \det(PL(U + \text{diag}(S))) = \det P \cdot \det L \cdot \det(U + \text{diag}(s))$$

$$|\det P| = 1, \quad |\det L| = 1 \text{ (all 1's on the diagonal)}$$

$$|\det(U + \text{diag}(s))| = \prod_i s_i \quad (\text{Because the elements on the diagonal are } 0+s_i)$$

$$\Rightarrow |\det W| = \prod_i |s_i| \rightarrow \log |\det W| = \sum_i \log |s_i|.$$

C: Efficient Inverse

for starters, we have :

$$\begin{aligned}
 Wz = b &\Rightarrow PL(U + \text{diag}(S))z = b \\
 &\xrightarrow{\times P^{-1}} L \underbrace{(U + \text{diag}(S))z}_y = P^\top b = \tilde{b} \\
 &\xrightarrow[\text{substitution}]{\text{forward}} y_i = \frac{\tilde{b}_i - \sum_{i < j} L_{ij} y_j}{L_{ii}} \Rightarrow \underbrace{(U + \text{diag}(S))z}_Q = y \\
 &\xrightarrow[\text{substitution}]{\text{backward}} z_i = \frac{y_i - \sum_{i > j} Q_{ij} z_j}{Q_{ii}}
 \end{aligned}$$

z is the desired $W^{-1}b$ and we have done it in $\mathcal{O}(C)$ for the permutation calculation, $\mathcal{O}(C^2)$ for the forward substitution, and $\mathcal{O}(C^2)$ for the backward substitution. The overall time complexity is $\mathcal{O}(C^2)$, which is more efficient than the case where we compute the explicit inverse which has time complexity of $\mathcal{O}(C^3)$.

Problem 2

Change of Variables Formula

From the standard change of variables formula we have that :

$$\begin{aligned}
 p_{z(t)}(z(t)) &= p_{z(0)}(z(0)) \left| \det \frac{\partial z(t)}{\partial z(0)} \right|^{-1} \\
 \log p_{z(t)}(z(t)) &= \log p_{z(0)}(z(0)) - \log \left| \det \frac{\partial z(t)}{\partial z(0)} \right|
 \end{aligned}$$

Differentiate w.r.t time

$$\frac{d \log p(z(t))}{dt} = - \frac{d \log \left| \det \frac{\partial z(t)}{\partial z(0)} \right|}{dt} = - \frac{1}{\left| \det \frac{\partial z(t)}{\partial z(0)} \right|} \cdot \frac{d \left| \det \frac{\partial z(t)}{\partial z(0)} \right|}{dt}$$

Apply Jacobi's formula

First we have to prove the Jacobi's formula. From laplace's expansion we have that :

$$\det(A) = \sum_i A_{ij} \text{adj}^\top(A)_{ij}$$

and that $\det(A)$ is a function of all of the elements of A , so :

$$\begin{aligned}
 \det(A) &= F(A_{11}, A_{12}, \dots, A_{nn}) \\
 d \det(A) &= \sum_i \sum_j \frac{\partial F}{\partial A_{ij}} dA_{ij} \\
 &\xrightarrow{\text{by fixing } i} \frac{d \det(A)}{dA_{ij}} = \frac{\partial \sum_k A_{ik} \text{adj}^\top(A)_{ik}}{\partial A_{ij}} = \sum_k \frac{\partial A_{ik} \text{adj}^\top(A)_{ik}}{\partial A_{ij}} \\
 \frac{\partial A_{ik} \text{adj}^\top(A)_{ik}}{\partial A_{ij}} &= \frac{\partial A_{ik}}{\partial A_{ij}} \text{adj}^\top(A)_{ik} + A_{ik} \frac{\partial \text{adj}^\top(A)_{ik}}{\partial A_{ij}}
 \end{aligned}$$

Because in the cofactor matrix we remove an entire row with fixed i , therefore all the elements A_{ij} are not in the cofactor matrix, therefore :

$$\begin{aligned} \frac{\partial A_{ik} \text{adj}^\top(A)_{ik}}{\partial A_{ij}} &= \frac{\partial A_{ik}}{\partial A_{ij}} \text{adj}^\top(A)_{ik} + A_{ik} \underbrace{\frac{\partial \text{adj}^\top(A)_{ik}}{\partial A_{ij}}}_0 \\ &= \underbrace{\frac{\partial A_{ik}}{\partial A_{ij}}}_{\delta_{jk} \text{ (Kronecker)}} \text{adj}^\top(A)_{ik} = \text{adj}^\top(A)_{ik} \delta_{jk} \end{aligned}$$

$$\begin{aligned} \frac{\partial \det(A)}{\partial A_{ij}} &= \sum_k \text{adj}^\top(A)_{ik} \delta_{jk} = \text{adj}^\top(A)_{ij} \\ \Rightarrow d \det(A) &= \sum_i \sum_j \text{adj}^\top(A)_{ij} dA_{ij} = \sum_j \sum_i \text{adj}(A)_{ji} dA_{ij} = \sum_j (\text{adj}(A) dA)_{jj} = \text{tr}(\text{adj}(A) dA) \end{aligned}$$

also we can write $\text{tr}(\text{adj}(A) dA)$ as :

$$\text{tr}(\text{adj}(A) dA) = \text{tr}(\det(A) A^{-1} dA) = \det(A) \text{tr}(A^{-1} dA)$$

Now that we have proved the Jacobi's formula, let's apply it to the result we got in the previous part:

$$\begin{aligned} \frac{\partial z(t)}{\partial z(0)} &= J(t) \\ \frac{d \log p(z(t))}{dt} &= -\frac{1}{|\det \frac{\partial z(t)}{\partial z(0)}|} \cdot \frac{d |\det \frac{\partial z(t)}{\partial z(0)}|}{dt} = -\frac{\det \frac{\partial z(t)}{\partial z(0)}}{\det \frac{\partial z(t)}{\partial z(0)}} \text{tr}(J(t)^{-1} \frac{\partial J(t)}{\partial t}) = -\text{tr}(J(t)^{-1} \frac{\partial J(t)}{\partial t}) \end{aligned}$$

Find the derivative of the Jacobian

$$\begin{aligned} \frac{dz}{dt} &= f_\theta(z(t), t) \\ \frac{d}{dt} \frac{\partial z(t)}{\partial z(0)} &= \frac{df_\theta(z(t), t)}{dz(0)} = \frac{\partial f_\theta(z(t), t)}{\partial z(t)} \frac{\partial z(t)}{\partial z(0)} = \frac{\partial f_\theta}{\partial z} J(t) \end{aligned}$$

Conclusion

$$\frac{d}{dt} \log p(z(t)) = -\text{tr}(J(t)^{-1} J(t) \frac{\partial f_\theta}{\partial z}) = -\text{tr}\left(\frac{\partial f_\theta}{\partial z}\right)$$

Problem 3 : Duality in Autoregressive Normalizing Flows

Theory of Masked Autoregressive Flow

Forward transformation for a single layer MAF is defined as :

$$\begin{aligned} f_\theta : z &\rightarrow x; z, x \in \mathbb{R}^D \\ x_i &= z_i \cdot \exp(\alpha_i(x_{<i})) + \mu_i(x_{<i}) \end{aligned}$$

Backward transformation is:

$$g_\theta : x \rightarrow z$$

$$z_i = \frac{x_i - \mu_i(x_{<i})}{\exp(\alpha_i(x_{<i}))}$$

Exact log-density is:

$$p_\theta(x) = p_z(g_\theta(x)) |\det \mathbf{J}_g(\mathbf{x})| = p_z(\mathbf{z}) |\det \mathbf{J}_f(\mathbf{z})|^{-1}$$

$$\mathbf{J}_f(\mathbf{z}) = \begin{bmatrix} \exp(\alpha_1(x_{<1})) & 0 & \dots & 0 \\ 0 & \exp(\alpha_2(x_{<2})) & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \exp(\alpha_D(x_{<D})) \end{bmatrix} \Rightarrow |\det \mathbf{J}_f(\mathbf{z})| = \left| \prod_{i=1}^D \exp(\alpha_i(x_{<i})) \right|$$

$$p_\theta(x) = \frac{p_z(\mathbf{z})}{\left| \prod_{i=1}^D \exp(\alpha_i(x_{<i})) \right|} \xrightarrow{\log} \log p_\theta(x) = \log p_z(\mathbf{z}) - \sum_{i=1}^D \alpha_i(x_{<i})$$

Now when we have a stacked MAF with K layers, we derive the full transformation in the following way: First consider output of each of the intermediate layers as $h^{(j)}$ where $h^{(j)} = f^{(j)}(h^{(j-1)})$, so we have :

$$h_i^{(j)} = h_i^{(j-1)} \exp(\alpha_i^{(j)}(h_{<i}^{(j)})) + \mu_i^{(j)}(h_{<i}^{(j)}), \quad h^{(0)} = z$$

$$\mathbf{x} = (f^{(K)} \circ f^{(K-1)} \circ \dots \circ f^{(1)})(\mathbf{z})$$

$$p_\theta(\mathbf{x}) = p_z(\mathbf{z}) |\det \mathbf{J}_{f^{(K)} \circ \dots \circ f^{(1)}}(\mathbf{z})|^{-1} \rightarrow \det \mathbf{J}_{f^{(K)} \circ \dots \circ f^{(1)}}(\mathbf{z}) = \prod_{j=1}^K \prod_{i=1}^D \exp(\alpha_i^{(j)}(h_{<i}^{(j)}))$$

$$\log p_\theta(\mathbf{x}) = \log p_z(\mathbf{z}) - \sum_{j=1}^K \sum_{i=1}^D \alpha_i^{(j)}(h_{<i}^{(j)})$$

Also, MAF enables fast density estimation due to the fact that the Jacobian is triangular and can be computed fast and that also in :

$$z_i = \frac{x_i - \mu_i(x_{<i})}{\exp(\alpha_i(x_{<i}))}$$

both α_i and μ_i depend on $x_{<i}$, which is already known when we process x , so each z_i can be computed efficiently. But sampling is another story; when we want to sample a vector $\mathbf{x}$ from the final distribution we have to calculate each x_i as :

$$x_i = z_i \cdot \exp(\alpha_i(x_{<i})) + \mu_i(x_{<i})$$

$$\rightarrow x_1 = z_1 \cdot \exp(\alpha_1(x_{<1})) + \mu_1(x_{<1}) \Rightarrow \begin{cases} \mu_2 = \mu_2(x_{<2}) \\ \alpha_2 = \alpha_2(x_{<2}) \end{cases}$$

$$\rightarrow x_2 = z_2 \cdot \exp(\alpha_2(x_{<2})) + \mu_2(x_{<2}) \Rightarrow \begin{cases} \mu_3 = \mu_3(x_{<3}) \\ \alpha_3 = \alpha_3(x_{<3}) \end{cases}$$

$\vdots$

$$\rightarrow x_D = z_D \exp(\alpha_D(x_{<D})) + \mu_D(x_{<D})$$

so for each x_i we first have to compute $x_{1:i-1}$, so for D elements it takes D autoregressive evaluations.

Theory of Inverse Autoregressive Flow

The forward transformation $f_\theta : z \rightarrow x$ is as follows :

$$x_d = z_d \cdot \exp(\alpha_d(z_{<d}) + \mu_d(z_{<d}))$$

and the backward $g_\theta : x \rightarrow z$:

$$z_d = \frac{x_d - \mu_d(z_{<d})}{\exp(\alpha_d(z_{<d}))}$$

The difference between MAF and IAF, is that IAF depends on the base distribution random variables instead of the target distribution.

Also it is well suited for sampling because z_i 's are known and we can use them to calculate μ_i and α_i which only depend on $z_{<i}$. After doing so we can easily calculate each vector x as:

$$x = z \odot \exp(\alpha) + \mu$$

For the density estimation, just like the sampling part in MAF, we have to sequentially calculate each z_i based on the backward transformation and each z_i depends on $z_{1:i-1}$. Therefore for D of them, we have to do D evaluations of the autoregressive model.

Equivalence Between MAF and IAF

First for the MAF we have :

$$f : u \rightarrow x$$

$$p_x(x) = p_u(u) |\det \mathbf{J}_f(u)|^{-1} \Rightarrow p_u(u) = p_x(f(u)) |\det \mathbf{J}_f(\mathbf{u})|$$

$$\begin{aligned} \mathbb{D}_{KL}(\pi_{\mathbf{x}}(\mathbf{x}) \parallel p_{\text{MAF}}(\mathbf{x})) &= \int \pi_{\mathbf{x}}(\mathbf{x}) \ln \frac{\pi_{\mathbf{x}}(\mathbf{x})}{p_{\text{MAF}}(\mathbf{x})} d\mathbf{x} \\ &= \int \pi_{\mathbf{x}}(\mathbf{x}) \ln \pi_{\mathbf{x}}(\mathbf{x}) d\mathbf{x} - \int \pi_{\mathbf{x}}(\mathbf{x}) \ln p_{\text{MAF}}(\mathbf{x}) d\mathbf{x} \\ &= \int \pi_{\mathbf{x}}(\mathbf{x}) \ln \pi_{\mathbf{x}}(\mathbf{x}) d\mathbf{x} - \int \pi_{\mathbf{x}}(\mathbf{x}) \ln \left(\pi_u(g(\mathbf{x})) |\det J_g(\mathbf{x})| \right) d\mathbf{x} \\ &= \int \pi_{\mathbf{x}}(\mathbf{x}) \ln \frac{\pi_{\mathbf{x}}(\mathbf{x})}{\pi_u(g(\mathbf{x})) |\det J_g(\mathbf{x})|} d\mathbf{x}. \end{aligned}$$

Now by the change of variable $x = f(u)$ and $d\mathbf{x} = \det J_f(\mathbf{u}) d\mathbf{u}$:

$$\begin{aligned} &= \int \pi_{\mathbf{x}}(f(\mathbf{u})) |\det J_f(\mathbf{u})| \ln \frac{\pi_{\mathbf{x}}(f(\mathbf{u}))}{\pi_u(\mathbf{u}) |\det J_f(\mathbf{u})|^{-1}} d\mathbf{u} \\ &= \int \underbrace{\pi_{\mathbf{x}}(f(\mathbf{u})) |\det J_f(\mathbf{u})|}_{p_u(\mathbf{u})} \ln \frac{\pi_{\mathbf{x}}(f(\mathbf{u})) |\det J_f(\mathbf{u})|}{\pi_u(\mathbf{u})} d\mathbf{u} \\ &= \int p_u(\mathbf{u}) \ln \frac{p_u(\mathbf{u})}{\pi_u(\mathbf{u})} d\mathbf{u} = \mathbb{D}_{KL}(p_u(\mathbf{u}) \parallel \pi_u(\mathbf{u})). \end{aligned}$$

Problem 4

(a) Deriving the General Loss

We start from the joint KL objective

$$\mathcal{L} = \iint \tilde{p}(x) p(z | x) \log \frac{\tilde{p}(x) p(z | x)}{q(x | z) q(z)} dz dx.$$

We are given the conditional-flow posterior

$$p(z | x) = \int \delta(z - F_x(u)) \mathcal{N}(u; 0, I) du, \quad H_x = F_x^{-1}.$$

■ (ai)

Substitute the delta-mixture form of $p(z | x)$ into $\mathcal{L}$ and use the sifting property of the Dirac delta:

$$\mathcal{L} = \iint \tilde{p}(x) \left[\int \delta(z - F_x(u)) \mathcal{N}(u; 0, I) du \right] \log \frac{\tilde{p}(x) \left[\int \delta(z - F_x(u')) \mathcal{N}(u'; 0, I) du' \right]}{q(x | z) q(z)} dz dx.$$

To make the delta usable inside the logarithm, we express the induced density of z given x by change-of-variables:

$$\int \delta(z - F_x(u')) \mathcal{N}(u'; 0, I) du' = \mathcal{N}(H_x(z); 0, I) \left| \det \frac{\partial H_x(z)}{\partial z} \right|.$$

Using $\left| \det \frac{\partial H_x(z)}{\partial z} \right| = \left| \det \frac{\partial F_x(u')}{\partial u'} \right|^{-1} \Big|_{u'=H_x(z)}$ we obtain

$$\int \delta(z - F_x(u')) \mathcal{N}(u'; 0, I) du' = \mathcal{N}(H_x(z); 0, I) \left| \det \left\{ \frac{\partial F_x(u')}{\partial u'} \right\} \right|^{-1} \Big|_{u'=H_x(z)}.$$

Now apply the sifting property to remove the z -integral:

$$\mathcal{L} = \iint \tilde{p}(x) \mathcal{N}(u; 0, I) \log \frac{\tilde{p}(x) \mathcal{N}(H_x(z); 0, I) \left| \det \left\{ \frac{\partial F_x(u')}{\partial u'} \right\} \right|^{-1} \Big|_{u'=H_x(z)}}{q(x | z) q(z)} \Big|_{z=F_x(u)} du dx.$$

Since $H_x(F_x(u)) = u$, this becomes

$$\mathcal{L} = \iint \tilde{p}(x) \mathcal{N}(u; 0, I) \log \frac{\tilde{p}(x) \mathcal{N}(u; 0, I)}{q(x | F_x(u)) q(F_x(u))} \left| \det \left\{ \frac{\partial F_x(u')}{\partial u'} \right\} \right|^{-1} \Big|_{u'=H_x(F_x(u))} du dx.$$

This matches the requested form (3).

Interpretation. The delta enforces the deterministic relation $z = F_x(u)$ under the approximate posterior. The remaining determinant factor appears because the delta over a nonlinear map induces a density over z via the inverse map.

■ (aii)

Let $v = F_x(u')$, so $u' = H_x(v)$ and

$$\left| \det \left\{ \frac{\partial u'}{\partial v} \right\} \right| = \left| \det \left\{ \frac{\partial F_x(u')}{\partial u'} \right\} \right|^{-1}.$$

From the previous step, the induced posterior density on z can be written as

$$p(z | x) = \int \delta(z - v) \mathcal{N}(H_x(v); 0, I) \left| \det \left\{ \frac{\partial H_x(v)}{\partial v} \right\} \right| dv.$$

Sifting with $\delta(z - v)$ gives

$$p(z | x) = \mathcal{N}(H_x(z); 0, I) \left| \det \left\{ \frac{\partial H_x(z)}{\partial z} \right\} \right| = \mathcal{N}(H_x(z); 0, I) \left| \det \left\{ \frac{\partial F_x(u')}{\partial u'} \right\} \right|_{u'=H_x(z)}^{-1}.$$

Now substitute $z = F_x(u)$ into $\mathcal{L}$ (the same sifting step as in (ai)), and use $q(z) = \mathcal{N}(z; 0, I)$:

$$\mathcal{L} = \iint \tilde{p}(x) \mathcal{N}(u; 0, I) \log \left(\frac{\tilde{p}(x) \mathcal{N}(u; 0, I)}{q(x | F_x(u)) \mathcal{N}(F_x(u); 0, I)} \left| \det \left\{ \frac{\partial F_x(u)}{\partial u} \right\} \right|^{-1} \right) du dx.$$

This is exactly (5).

■ (aiii)

(i) Determinant in the denominator. The posterior is defined by pushing forward a base variable $u \sim \mathcal{N}(0, I)$ through $z = F_x(u)$. For an invertible map, densities transform as

$$p(z | x) = p(u) \left| \det \frac{\partial u}{\partial z} \right| = p(u) \left| \det \frac{\partial F_x(u)}{\partial u} \right|^{-1}, \quad u = H_x(z).$$

Thus the determinant is in the denominator because we are using $\det(\partial z / \partial u)$, while the density needs $\det(\partial u / \partial z) = \det(\partial z / \partial u)^{-1}$.

(ii) Expressivity over diagonal Gaussians. A diagonal Gaussian posterior has the restricted form

$$p_{\text{diag}}(z | x) = \mathcal{N}(z; \mu_\phi(x), \text{diag}(\sigma_\phi^2(x))),$$

whose density level sets are axis-aligned ellipsoids and cannot represent complex multimodality or warped correlation structures. In contrast, a conditional flow posterior is the pushforward of a simple base:

$$u \sim \mathcal{N}(0, I), \quad z = F_x(u), \quad p(z | x) = \mathcal{N}(u; 0, I) \left| \det \frac{\partial F_x(u)}{\partial u} \right|^{-1}.$$

Because F_x can be an expressive, invertible neural network conditioned on x , it can induce rich families of posteriors with non-linear correlations, non-Gaussian tails, and (with sufficient depth) highly flexible shapes, strictly containing the diagonal-Gaussian family as a special case (e.g. when F_x is affine and diagonal in u).

(b) Recovering Flow-Based Models

We choose

$$F_x(u) = F(\sigma u + x), \quad q(x | z) = \mathcal{N}(x; F^{-1}(z), \sigma I), \quad \sigma > 0,$$

where F is an unconditional invertible flow.

■ (bi)

Let $w = \sigma u + x$. Then $F_x(u) = F(w)$ and $F^{-1}(F_x(u)) = F^{-1}(F(w)) = w$. Therefore,

$$\log q(x | F_x(u)) = \log \mathcal{N}(x; F^{-1}(F_x(u)), \sigma^2 I) = \log \mathcal{N}(x; w, \sigma^2 I).$$

Since $w = x + \sigma u$, we have $x - w = -\sigma u$ and $\|x - w\|^2 = \sigma^2 \|u\|^2$. Hence

$$\log q(x | F_x(u)) = -\frac{d}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \|x - w\|^2 = -\frac{d}{2} \log(2\pi\sigma^2) - \frac{1}{2} \|u\|^2,$$

which is (7).

■ (bii)

We have $F_x(u) = F(w)$ with $w = \sigma u + x$ and $\frac{\partial w}{\partial u} = \sigma I$. By the chain rule,

$$\frac{\partial F_x(u)}{\partial u} = \frac{\partial F(w)}{\partial w} \frac{\partial w}{\partial u} = \frac{\partial F(w)}{\partial w} (\sigma I).$$

Taking determinants,

$$\left| \det \frac{\partial F_x(u)}{\partial u} \right| = \left| \det \frac{\partial F(w)}{\partial w} \right| \cdot |\det(\sigma I)| = \left| \det \frac{\partial F(w)}{\partial w} \right| \cdot \sigma^d.$$

Taking logs yields

$$\log \left| \det \frac{\partial F_x(u)}{\partial u} \right| = d \log \sigma + \log \left| \det \frac{\partial F(w)}{\partial w} \right|,$$

which is (8).

■ (biii)

Start from (5):

$$\mathcal{L} = \iint \tilde{p}(x) \mathcal{N}(u; 0, I) \log \left(\frac{\tilde{p}(x) \mathcal{N}(u; 0, I)}{q(x | F_x(u)) \mathcal{N}(F_x(u); 0, I)} \left| \det \frac{\partial F_x(u)}{\partial u} \right|^{-1} \right) du dx.$$

Expand the log:

$$\mathcal{L} = \mathbb{E}_{x \sim \tilde{p}, u \sim \mathcal{N}(0, I)} \left[\log \tilde{p}(x) + \log \mathcal{N}(u; 0, I) - \log q(x | F_x(u)) - \log \mathcal{N}(F_x(u); 0, I) - \log \left| \det \frac{\partial F_x(u)}{\partial u} \right| \right].$$

Now insert (7):

$$- \log q(x | F_x(u)) = \frac{d}{2} \log(2\pi\sigma^2) + \frac{1}{2} \|u\|^2,$$

which depends only on σ and u , and contains *no trainable parameters in F nor dependence on x beyond constants*. Also combine the base and reconstruction Gaussian terms:

$$\log \mathcal{N}(u; 0, I) + \frac{1}{2} \|u\|^2 = -\frac{d}{2} \log(2\pi),$$

a constant. Therefore, up to constants independent of F , the loss reduces to

$$\mathcal{L} = \mathbb{E}_{x \sim \tilde{p}, u \sim \mathcal{N}(0, I)} \left[-\log \mathcal{N}(F_x(u); 0, I) - \log \left| \det \frac{\partial F_x(u)}{\partial u} \right| \right] + \text{const.}$$

Using $-\log \mathcal{N}(z; 0, I) = \frac{1}{2} \|z\|^2 + \frac{d}{2} \log(2\pi)$ and dropping constants:

$$\mathcal{L} = \mathbb{E}_{x \sim \tilde{p}, u \sim \mathcal{N}(0, I)} \left[\frac{1}{2} \|F_x(u)\|^2 - \log \left| \det \frac{\partial F_x(u)}{\partial u} \right| \right] + \text{const.}$$

Finally substitute $F_x(u) = F(\sigma u + x)$:

$$\mathcal{L} = \mathbb{E}_{x \sim \tilde{p}, u \sim \mathcal{N}(0, I)} \left[\frac{1}{2} \|F(\sigma u + x)\|^2 - \log \left| \det \frac{\partial F(\sigma u + x)}{\partial u} \right| \right] + \text{const},$$

which is (9).

Conclusion (flow-based training with dequantization noise). The objective is exactly the standard maximum-likelihood flow loss on inputs perturbed by noise $x \mapsto x + \sigma u$ (often called input dequantization), because it matches the negative log-likelihood of the flow applied to the noisy data, up to constants.

Problem 5: VP-NFs are not universal

Part (a)

In this part we introduce an event E that satisfies the desired condition. Note that k is chosen such that the integration of $p(x, y)$ over x, y is equal to 1.

$$A = \implies \exists \varepsilon > 0 \quad \forall (x, y) \in \mathbb{R}^2 : \quad p_\theta(x, y) < 0.9 - \varepsilon$$

$$\bar{A} = B \setminus A = B, \quad p(\bar{A}) = p(B) = 0.9, \quad |\bar{A}| = 1$$

$$p_\theta(\bar{A}) = \int_{(x,y) \in \bar{A}} p_\theta(x, y) dx dy < \int_{(x,y) \in \bar{A}} (0.9 - \varepsilon) dx dy = (0.9 - \varepsilon)|\bar{A}| = 0.9 - \varepsilon$$

Define $E := \bar{A}$. Then

$$p(E) - p_\theta(E) = \varepsilon > 0 \implies |p(E) - p_\theta(E)| > 0$$

Using Pinsker's inequality,

$$\sqrt{\frac{1}{2} D_{\text{KL}}(p_\theta \| p)} > 0 \implies D_{\text{KL}}(p_\theta \| p) > 0.$$

Part (b)

According to our assumption, the Jacobian determinant is constant. We denote this constant by J .

$$(x, y) = f(z), \quad z \sim \mathcal{N}(0, I)$$

$$p_\theta(x, y) = p(z) \left| \det \frac{\partial f_\theta(z)}{\partial z} \right|^{-1}, \quad J := \left| \det \frac{\partial f_\theta(z)}{\partial z} \right|^{-1}$$

Define the set

$$A := \{(x, y) \in \mathbb{R}^2 : p(z = f_\theta(x, y)) J \geq 0.9 - \varepsilon\}.$$

Based on the latent representation of A , define

$$C := \{z \in \mathbb{R}^2 : p(z) J \geq 0.9 - \varepsilon\}.$$

Since

$$p(z) = \frac{1}{2\pi} e^{-\|z\|^2/2},$$

we obtain

$$\begin{aligned} p(z) J \geq 0.9 - \varepsilon &\iff \frac{J}{2\pi} e^{-\|z\|^2/2} \geq 0.9 - \varepsilon \\ \implies \|z\|^2 &\leq 2 \log \left(\frac{J}{2\pi(0.9 - \varepsilon)} \right), \quad J \geq 2\pi(0.9 - \varepsilon) \end{aligned}$$

Thus C is a centered 2D ball with

$$\text{vol}(C) = \pi R^2 = 2\pi \log \left(\frac{J}{2\pi(0.9 - \varepsilon)} \right).$$

Using the change-of-variables formula,

$$\text{vol}(A) = \frac{\text{vol}(C)}{J}.$$

Hence,

$$\text{vol}(A) = \frac{2\pi}{J} \log\left(\frac{J}{2\pi(0.9 - \varepsilon)}\right) \leq \frac{1}{e(0.9 - \varepsilon)},$$

where we used $\log x \leq \frac{x}{e}$ for all $x > 0$.

Part (c)

As in part (a), consider the event $E := \bar{A}$.

$$\bar{A} \cup A = B \implies |\bar{A}| + |A| \geq |B| = 1 \implies |\bar{A}| \geq 1 - |A|$$

$$|p(E) - p_\theta(E)| = |p(\bar{A}) - p_\theta(\bar{A})| = |0.9|\bar{A}| - p_\theta(\bar{A})|$$

$$\geq |0.9|\bar{A}| - (0.9 - \varepsilon)|\bar{A}|| = |\bar{A}|\varepsilon$$

$$\geq (1 - |A|)\varepsilon \geq \left(1 - \frac{1}{e(0.9 - \varepsilon)}\right)\varepsilon > 0$$

Again, by Pinsker's inequality,

$$\sqrt{\frac{1}{2}D_{\text{KL}}(p_\theta\|p)} > 0 \implies D_{\text{KL}}(p_\theta\|p) > 0.$$

Part (d)

From parts (a) and (c), we showed that in this special case there exists a strictly positive distance between p and p_θ when using normalizing flows with constant absolute Jacobian determinant.

This implies that, when KL-divergence is used as the distance metric, no volume-preserving normalizing flow can approximate the true distribution arbitrarily well.